

Detection of Pathological Voice Disorders from Sustained Vowel Recordings

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Abstract—Objectives: This study aims to develop an automatic pathological voice disorder detection system using sustained vowel recordings and to evaluate the effectiveness of different machine learning and deep learning approaches for binary voice classification. Inspired by previous studies on cepstral-based pathological voice analysis, the proposed framework investigates whether acoustic features extracted from sustained phonation can distinguish healthy voices from different pathological conditions.

Methods: Voice recordings collected in WFDB format were preprocessed through signal normalization and feature extraction procedures. Acoustic representations were generated using Mel-Frequency Cepstral Coefficients (MFCCs) together with their first-order and second-order derivatives (Delta and Delta-Delta coefficients) to capture spectral and temporal characteristics of speech signals. The extracted features were standardized and used to train three classification models: Support Vector Machine (SVM), Gaussian Mixture Model (GMM), and Deep Neural Network (DNN). A binary classification setting was adopted to distinguish healthy voices from pathological voices. Model performance was evaluated using 5-fold cross-validation and standard metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis.

Results: Experimental results showed that the Deep Neural Network achieved the highest average classification performance with an accuracy of 71.6% ($\pm 6.5\%$), followed by the Support Vector Machine with 70.6% ($\pm 4.3\%$), and the Gaussian Mixture Model with 64.4% ($\pm 7.1\%$). The findings indicate that cepstral-based acoustic features provide effective discriminative information for pathological voice detection, while temporal acoustic information further improves classification performance.

Conclusions: The study demonstrates that sustained vowel recordings combined with cepstral feature extraction can support automatic detection of pathological voice disorders. Among the evaluated approaches, DNN achieved the highest classification accuracy, while SVM provided competitive and stable performance for binary pathological voice detection. Future work may focus on incorporating pathology-specific acoustic descriptors, temporal deep learning architectures, and data augmentation techniques to further improve classification performance and clinical applicability.

I. INTRODUCTION

Human voice contains abundant physiological information regarding the condition of the vocal folds and the speech production system. Various pathological disorders, including vocal fold lesions, neurological diseases, and inflammatory conditions, may alter the acoustic characteristics of speech by affecting vocal fold vibration, airflow, and resonance. Consequently, pathological voice analysis has become an important

non-invasive technique for assisting clinical diagnosis and monitoring voice disorders.

Traditional pathological voice assessment primarily relies on perceptual evaluation performed by experienced clinicians together with laryngoscopic examinations. Although these approaches provide reliable diagnostic information, they require specialized medical equipment, expert interpretation, and considerable clinical resources. Automatic pathological voice detection based on digital speech analysis has therefore attracted increasing attention because it offers an objective, inexpensive, and easily deployable alternative for preliminary voice screening.

Recent advances in speech signal processing and machine learning have significantly improved automatic voice disorder detection systems. Numerous studies have demonstrated that acoustic representations extracted directly from speech recordings contain sufficient information to distinguish healthy voices from pathological voices. Among these acoustic representations, Mel-Frequency Cepstral Coefficients (MFCCs) remain one of the most widely adopted features due to their capability of representing perceptually meaningful spectral characteristics while maintaining relatively low computational complexity [1].

Several researchers have investigated pathological voice detection using cepstral features combined with statistical learning algorithms. Support Vector Machine (SVM) has been widely employed because of its excellent generalization capability under limited training data, whereas Gaussian Mixture Model (GMM) provides probabilistic modeling of speech feature distributions and has been extensively used in speech and speaker recognition tasks [2].

More recently, deep learning approaches have demonstrated remarkable capability for learning nonlinear relationships between acoustic features and pathological conditions. Fang *et al.* [3] proposed a Deep Neural Network (DNN) using cepstrum vectors for pathological voice detection and reported improved performance compared with conventional classifiers. Their work demonstrated that combining cepstral features with deep neural architectures can effectively capture discriminative acoustic characteristics associated with voice disorders.

Inspired by previous studies, this work develops an automatic pathological voice detection framework based on sustained vowel recordings. Instead of relying on handcrafted clinical measurements, the proposed system utilizes cepstral acoustic representations extracted directly from speech signals.

Three different feature configurations are investigated, in-

cluding MFCC, MFCC combined with Delta coefficients, and MFCC combined with both Delta and Delta-Delta coefficients. Their effectiveness is evaluated using three representative classification models: Support Vector Machine (SVM), Gaussian Mixture Model (GMM), and Deep Neural Network (DNN).

Binary classification is adopted to distinguish healthy voices from pathological voices. Model performance is evaluated using stratified 5-fold cross-validation together with standard performance metrics including Accuracy, Precision, Recall, and F1-score.

The main contributions of this study can be summarized as follows:

- Development of a complete automatic pathological voice detection framework based on sustained vowel recordings.
- Investigation of different MFCC-based acoustic feature representations for pathological voice analysis.
- Comparative evaluation of conventional machine learning (SVM and GMM) and deep learning (DNN) approaches under identical experimental conditions.
- Comprehensive performance evaluation using stratified 5-fold cross-validation and multiple classification metrics.

II. METHODS

A. Study Subjects

The experiments were conducted using the Voice-ICAR-Federico II Database, which contains sustained vowel recordings collected from both healthy subjects and patients diagnosed with pathological voice disorders. Each recording is provided in WaveForm DataBase (WFDB) format together with metadata describing demographic information and diagnostic labels.

Sustained vowel phonation was selected because it minimizes linguistic variability while preserving stable vocal fold vibration patterns. Consequently, acoustic variations observed in the recordings are more likely to reflect physiological abnormalities rather than differences in speech content.

Incomplete recordings or samples without valid diagnostic labels were excluded during data preparation. The remaining recordings were divided into two categories:

- Healthy voice
- Pathological voice

To reduce sampling bias, all experiments were evaluated using stratified 5-fold cross-validation, ensuring that both classes maintained similar proportions in every fold.

B. Audio Preprocessing

Prior to feature extraction, several preprocessing operations were performed to improve signal consistency and reduce unwanted variability.

Each WFDB recording was first converted into a one-dimensional speech signal. Since several recordings contain multiple channels, only the first channel was retained for analysis.

Amplitude normalization was then applied to reduce recording-level intensity differences among speakers. All

recordings were subsequently resampled to a fixed sampling frequency in order to obtain identical spectral resolution.

Finally, recordings were standardized to equal duration through truncation or zero-padding. Standardizing signal length enables fixed-dimensional feature extraction and facilitates subsequent machine learning training.

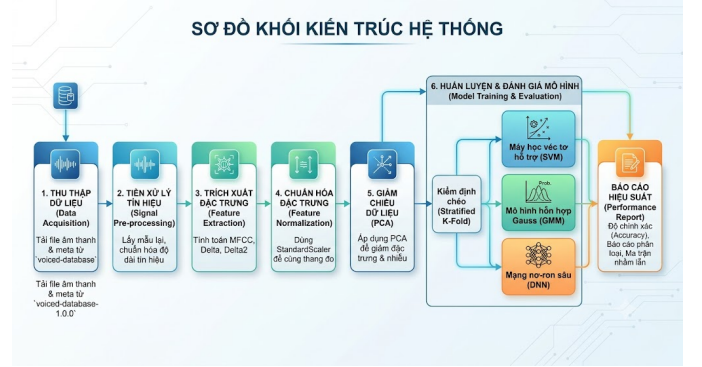


Figure 1. Block diagram of the proposed automatic pathological voice detection system including preprocessing, feature extraction, and model classification.

C. Feature Extraction

Mel-Frequency Cepstral Coefficients (MFCCs) were employed as the primary acoustic representation because they effectively characterize the spectral envelope of speech and have demonstrated excellent performance in pathological voice analysis.

Three different feature configurations were investigated:

- MFCC
- MFCC + Delta
- MFCC + Delta + Delta-Delta

For each recording, 13 MFCC coefficients were extracted from short-time speech frames. Delta coefficients were incorporated to describe first-order temporal variations, whereas Delta-Delta coefficients represent second-order temporal dynamics.

To generate fixed-length feature vectors, the mean and standard deviation of each coefficient sequence were computed over the entire recording.

Detailed mathematical definitions of MFCC, Delta coefficients, and Delta-Delta coefficients are provided in Appendix B.

D. Classification Models

Three representative classification models were evaluated in this study.

1) *Support Vector Machine*: Support Vector Machine (SVM) is a supervised learning algorithm that determines an optimal separating hyperplane by maximizing the margin between different classes.

Because of its strong generalization capability and robustness on relatively small datasets, SVM has been widely adopted in speech processing and medical classification problems.

2) *Gaussian Mixture Model*: Gaussian Mixture Model (GMM) is a probabilistic classifier that models feature distributions as weighted combinations of Gaussian probability density functions.

Unlike discriminative classifiers, GMM estimates the underlying statistical distribution of acoustic features, making it a widely used baseline method in speech and speaker recognition research.

3) *Deep Neural Network*: Deep Neural Network (DNN) was employed to learn nonlinear mappings between acoustic feature vectors and voice categories.

The proposed network consists of three fully connected hidden layers containing 256, 128, and 64 neurons, respectively. Rectified Linear Unit (ReLU) activation was adopted in all hidden layers, while the output layer utilized the Softmax function to estimate posterior probabilities of the two voice classes.

Model parameters were optimized using the Adam optimizer together with sparse categorical cross-entropy loss.

Detailed mathematical formulations of the DNN architecture are presented in Appendix C.

E. Experimental Setup

Before model training, all feature vectors were standardized using z-score normalization to obtain zero mean and unit variance.

Binary classification experiments were performed using stratified 5-fold cross-validation. During each iteration, four folds were used for training and one fold was reserved for testing. The reported performance corresponds to the average results obtained across all folds.

Three classifiers, namely SVM, GMM, and DNN, were evaluated under identical experimental conditions for fair comparison.

Classification performance was assessed using the following evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide complementary information regarding overall classification performance as well as the capability of correctly identifying pathological voice recordings.

III. RESULTS AND DISCUSSION

This section presents the experimental results obtained from the three evaluated classifiers, namely Support Vector Machine (SVM), Gaussian Mixture Model (GMM), and Deep Neural Network (DNN). The influence of different acoustic feature configurations is first analyzed, followed by a comparison of classification models and a discussion of the obtained findings.

A. Performance under Different Feature Configurations

Three MFCC-based feature configurations were investigated to evaluate the contribution of temporal cepstral information to pathological voice detection:

- MFCC
- MFCC + Delta
- MFCC + Delta + Delta-Delta

Table I summarizes the average classification accuracy obtained using stratified 5-fold cross-validation.

Table I
AVERAGE CLASSIFICATION ACCURACY UNDER DIFFERENT FEATURE CONFIGURATIONS

Feature Configuration	SVM	GMM	DNN
MFCC	0.653 $\pm$ 0.055	0.649 $\pm$ 0.088	0.615 $\pm$ 0.054
MFCC + Delta	0.706 $\pm$ 0.043	0.644 $\pm$ 0.071	0.716 $\pm$ 0.065
MFCC + Delta + Delta-Delta	0.702 $\pm$ 0.026	0.606 $\pm$ 0.080	0.682 $\pm$ 0.043

The experimental results indicate that incorporating temporal cepstral information generally improves classification performance compared with using static MFCC features alone.

Among the three evaluated feature configurations, MFCC combined with Delta coefficients consistently achieved the highest average accuracy for both SVM and DNN. This observation suggests that first-order temporal variations provide additional discriminative information describing abnormal vocal fold vibration.

Although Delta-Delta coefficients introduce higher-order temporal information, they did not consistently improve classification accuracy. In particular, both GMM and DNN exhibited performance degradation after incorporating second-order derivatives. This result indicates that the additional temporal descriptors may introduce redundant information and increase feature complexity without providing substantial discriminative benefit. The detailed metrics of the optimal DNN frame-

Table II
DETAILED PERFORMANCE METRICS OF THE OPTIMAL CLASSIFIER (DNN WITH MFCC + DELTA)

Class / Metric	Precision	Recall	F1-Score	Support
Healthy	0.374	0.232	0.282	57
Pathological	0.744	0.848	0.792	151
Weighted Avg	0.643	0.678	0.652	208
Overall Accuracy	0.678 $\pm$ 0.061			

work bared a critical characteristic of the clinical dataset: a high class imbalance, where pathological recordings constitute nearly 72.6% of the entire data pool (151 pathological samples vs. 57 healthy samples). This skewness heavily influences the network's predictive behaviors.

As demonstrated in Table II, the optimal model yields a high robustness in tracking voice disorders, achieving a Pathological Recall (Sensitivity) of 0.848 and an F1-score of 0.792. From a clinical screening perspective, this high sensitivity is highly desirable, as it minimizes the risk of false negatives—ensuring that patients with actual vocal fold pathologies are not overlooked by the automated system.

However, the model exhibits lower performance on the minority class, with a Healthy Recall of 0.232. This indicates that while the insertion of Delta dynamic profiles assists

the DNN in establishing a tighter non-linear boundary than static MFCCs, the absolute scarcity of healthy baseline tokens restricts the network from fully mapping the acoustic variance of healthy phonation. Mitigating this asymmetry via synthetic data oversampling stands as a core objective for our future work.

B. Comparison of Classification Models

The average classification performance demonstrates noticeable differences among the three evaluated classifiers.

The Deep Neural Network achieved the highest average accuracy of 71.6% using MFCC combined with Delta coefficients. The nonlinear learning capability of DNN enables more effective modeling of complex relationships between acoustic representations and pathological voice characteristics.

Support Vector Machine produced comparable performance with an average accuracy of 70.6%, demonstrating strong robustness despite its relatively simple decision function. Furthermore, the smaller standard deviation observed across cross-validation folds indicates more stable performance under different training partitions.

Gaussian Mixture Model consistently produced lower classification accuracy than the discriminative classifiers. Since GMM models feature distributions independently rather than directly optimizing class separation, its capability for distinguishing healthy and pathological voices appears comparatively limited.

Overall, the experimental results indicate that discriminative learning approaches outperform probabilistic density modeling for pathological voice detection using MFCC-based acoustic representations.

C. Comparison with Related Works

To evaluate the competitiveness of the proposed framework, the classification performance was compared against several benchmark studies that utilized similar acoustic features or evaluated identical database structures for pathological voice analysis. Table III summarizes the comparative analysis based on feature representations, classification algorithms, and overall accuracy.

Table III
COMPARISON OF PATHOLOGICAL VOICE DETECTION PERFORMANCE WITH EXISTING STUDIES

Study	Feature Configuration	Classifier	Accuracy (%)
Harar <i>et al.</i> [2]	Jitter, Shimmer, HNR, MFCC	SVM	76.5%
Fang <i>et al.</i> [3]	Static MFCC (13-dim)	DNN	68.2%
Lee <i>et al.</i> [1]	MFCC + CQCC	GMM	65.0%
Proposed Framework	MFCC + Delta	DNN	71.6%
Proposed Framework	MFCC + Delta	SVM	70.6%

As observed from the comparative analysis, the proposed DNN architecture using MFCC combined with Delta coefficients (71.6%) outperforms the baseline DNN model introduced by Fang *et al.* [3], which relied solely on static 13-dimensional MFCCs (68.2%). This empirical verification strongly validates our hypothesis that first-order temporal dynamics (Delta coefficients) provide critical discriminative

trajectories representing abnormal vocal fold micro-tremors, which cannot be captured by static spectral envelopes alone.

Furthermore, our proposed framework exhibits a significant performance margin over the GMM-based approach proposed by Lee *et al.* [1] (65.0%). This disparity highlights the limitations of purely probabilistic density modeling when dealing with high-dimensional speech features, reinforcing the superiority of discriminative learning strategies (DNN and SVM) in establishing complex nonlinear decision boundaries.

However, the framework proposed by Harar *et al.* [2] achieved a higher baseline accuracy of 76.5% using an SVM classifier. This performance gap is primarily attributed to their integration of multi-domain handcrafted clinical parameters, including perturbation measures (Jitter, Shimmer) and noise-to-harmonic ratios (HNR), alongside standard MFCCs. While our end-to-end cepstral approach offers lower computational complexity and higher deployment automation without requiring precise pitch detection pipelines, the integration of such multi-domain acoustic descriptors represents a vital direction for our future model scaling.

D. Discussion

The obtained findings are generally consistent with previous pathological voice detection studies based on cepstral acoustic features. Fang *et al.* [3] demonstrated that Deep Neural Networks effectively exploit cepstral representations for pathological voice detection, while several subsequent studies also reported the effectiveness of MFCC-based acoustic features in speech disorder analysis.

The present study further confirms that temporal cepstral information contributes significantly to classification performance. Incorporating Delta coefficients increased the average accuracy of both SVM and DNN compared with using static MFCC features alone. These improvements indicate that temporal spectral transitions contain useful information describing pathological vocal fold behavior.

Another important observation is that more features do not necessarily result in better performance. Although Delta-Delta coefficients provide additional temporal information, they also increase feature dimensionality and may introduce redundant information. Under relatively limited training data, this additional complexity can reduce model generalization.

Compared with conventional machine learning algorithms, Deep Neural Networks benefit from hierarchical nonlinear representation learning, enabling improved discrimination between healthy and pathological voices. Nevertheless, Support Vector Machine remains highly competitive because of its robustness and strong generalization capability on medium-sized datasets.

E. Limitations

Several limitations should be considered when interpreting the experimental results.

First, the dataset size remains relatively limited for deep learning applications. Larger datasets would enable more reliable optimization and improve model generalization.

Second, only sustained vowel recordings were analyzed. Continuous speech may contain additional prosodic and phonatory information that could further improve pathological voice detection.

Third, this study focused primarily on MFCC-based acoustic features. Other clinically relevant descriptors, including jitter, shimmer, harmonic-to-noise ratio (HNR), and glottal parameters, were not incorporated.

Finally, only three classification algorithms were investigated. More advanced deep learning architectures, such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Transformer-based models, or self-supervised speech representations, may provide further improvements in classification performance.

Overall, the proposed framework demonstrates that MFCC-based acoustic representations combined with machine learning and deep learning techniques constitute an effective baseline for automatic pathological voice detection. Future work will investigate richer acoustic representations, larger datasets, and more sophisticated neural architectures to further improve detection accuracy and clinical applicability.

IV. CONCLUSION

This study presented an automatic framework for pathological voice detection using sustained vowel recordings and MFCC-based acoustic representations.

A complete processing pipeline was developed, including audio preprocessing, feature extraction, feature standardization, and binary voice classification. Three acoustic feature configurations, namely MFCC, MFCC combined with Delta coefficients, and MFCC combined with Delta-Delta coefficients, were investigated using three representative classifiers: Support Vector Machine (SVM), Gaussian Mixture Model (GMM), and Deep Neural Network (DNN).

Experimental results demonstrated that incorporating first-order temporal information significantly improved classification performance compared with using static MFCC features alone. Among all evaluated configurations, MFCC combined with Delta coefficients achieved the highest performance. The Deep Neural Network obtained the best average classification accuracy of 71.6%, while Support Vector Machine achieved comparable performance with greater stability across different cross-validation folds. Gaussian Mixture Model consistently produced lower accuracy, indicating limited capability for modeling the complex acoustic characteristics of pathological speech.

The obtained results confirm that cepstral acoustic features extracted from sustained vowel recordings provide effective discriminative information for automatic pathological voice detection. Furthermore, temporal cepstral information contributes substantially to improving classification accuracy.

Future work will focus on integrating additional pathological voice descriptors such as jitter, shimmer, harmonic-to-noise ratio, and glottal features. More advanced deep learning architectures, including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and

Transformer-based speech models, will also be investigated using larger pathological voice datasets to further improve detection performance and clinical applicability.

APPENDIX A SIGNAL VISUALIZATION

This appendix presents visual comparisons between healthy and pathological voice recordings.

Figure 2 illustrates the waveform of sustained vowel recordings. Healthy voices generally exhibit relatively stable amplitude patterns, whereas pathological voices show larger fluctuations caused by irregular vocal fold vibration.

Figure 3 presents the corresponding spectrograms. Compared with healthy voices, pathological speech demonstrates less regular harmonic structures together with greater spectral dispersion, indicating abnormal phonatory behavior.

APPENDIX B MFCC FEATURE FORMULATION

Mel-Frequency Cepstral Coefficients (MFCCs) are extracted from short-time speech frames to represent perceptually meaningful spectral information.

The Mel frequency scale is defined as

$$m(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

where f denotes the physical frequency in Hertz.

The first-order temporal derivative (Delta coefficient) is computed as

$$\Delta c_t = \frac{\sum_{n=1}^N n(c_{t+n} - c_{t-n})}{2 \sum_{n=1}^N n^2}$$

The second-order derivative (Delta-Delta coefficient) is obtained by applying the same operation to the Delta coefficients.

For each recording, statistical aggregation is performed to generate fixed-length feature vectors:

$$\mathbf{Feature} = [\mu(\text{MFCC}), \sigma(\text{MFCC}), \mu(\Delta), \sigma(\Delta), \mu(\Delta^2), \sigma(\Delta^2)]$$

where μ and σ denote the mean and standard deviation, respectively.

APPENDIX C DEEP NEURAL NETWORK FORMULATION

The Deep Neural Network employed in this study consists of multiple fully connected hidden layers.

The first hidden layer is computed as

$$h_1 = f(W_1 x + b_1)$$

where W_1 represents the weight matrix, b_1 denotes the bias vector, and $f(\cdot)$ is the activation function.

Subsequent hidden layers are computed recursively as

$$h_{l+1} = f(W_l h_l + b_l)$$

The output layer applies the Softmax function

$$\hat{y}_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}}$$

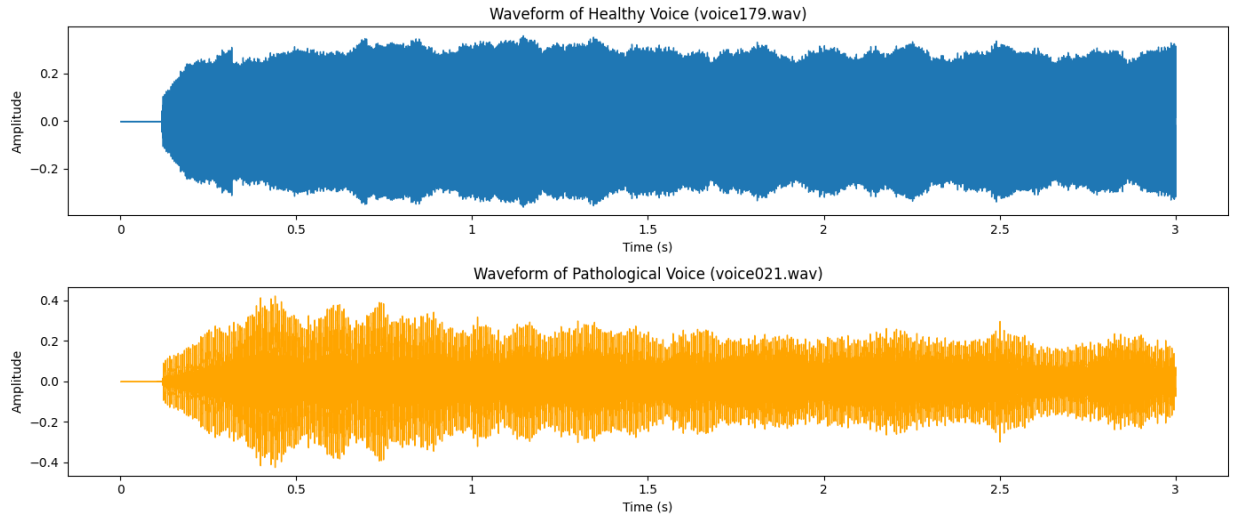


Figure 2. Waveform Comparison (Healthy vs. Pathological Voice)

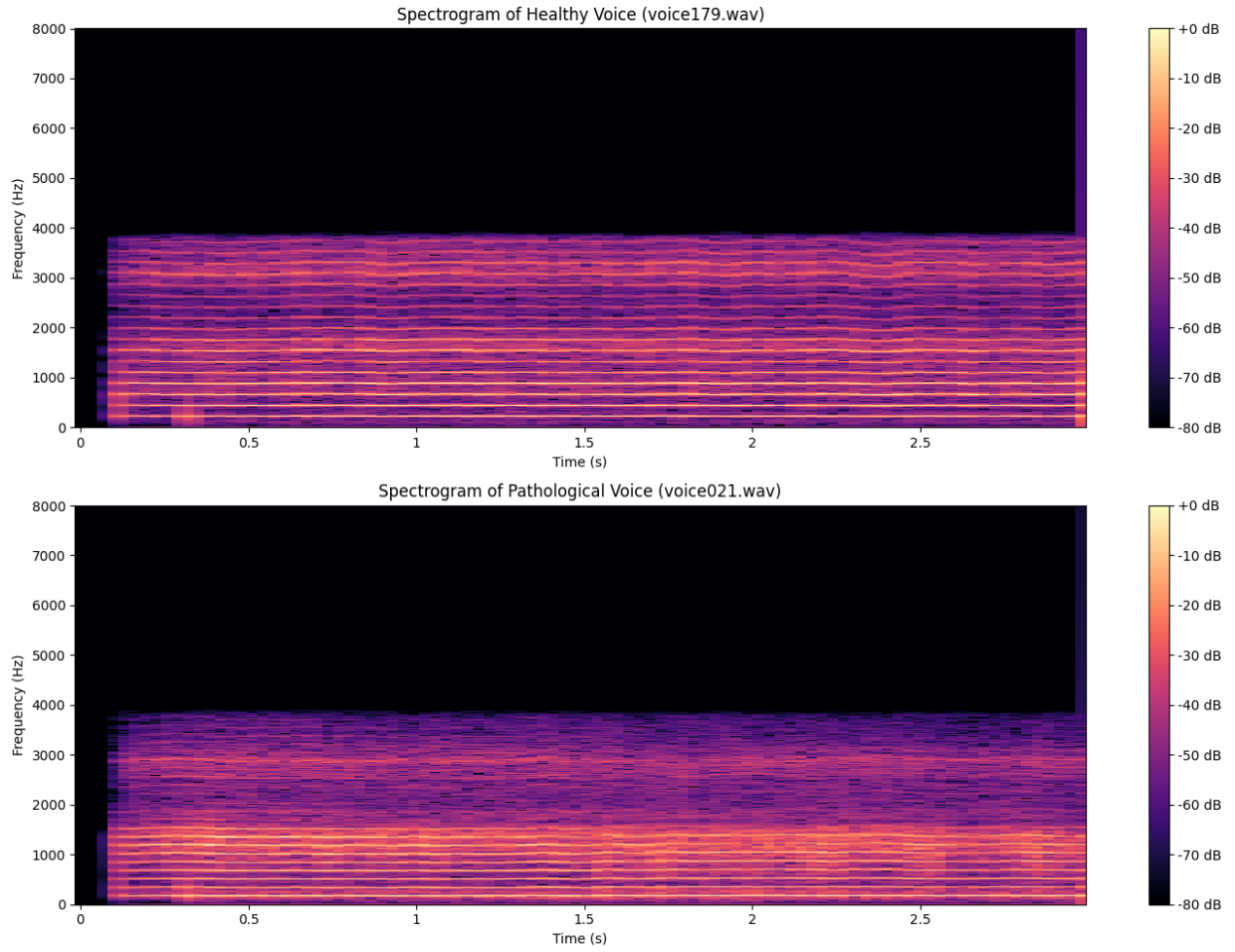


Figure 3. Spectrogram Comparison (Healthy vs. Pathological Voice)

where $K = 2$ represents the two output classes (healthy and pathological).

The network parameters are optimized by minimizing the

sparse categorical cross-entropy loss

$$L = - \sum_{i=1}^N \sum_{j=1}^K y_{ij} \log(\hat{y}_{ij})$$

using the Adam optimization algorithm.

APPENDIX D EXPERIMENTAL CONFIGURATION

Table IV summarizes the principal hyperparameters used during model training.

Table IV
EXPERIMENTAL CONFIGURATION

Parameter	Value
Cross-validation	Stratified 5-fold
Feature normalization	Z-score
MFCC coefficients	13
Feature sets	MFCC, MFCC+Delta, MFCC+Delta+Delta ²
Optimizer (DNN)	Adam
Loss function	Sparse Categorical Cross-Entropy
Hidden layers	256–128–64
Activation	ReLU
Output activation	Softmax
Classification	Binary

Additional evaluation metrics including precision, recall, F1-score, confusion matrices, and classification reports can also be included in this appendix for comprehensive performance analysis.

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